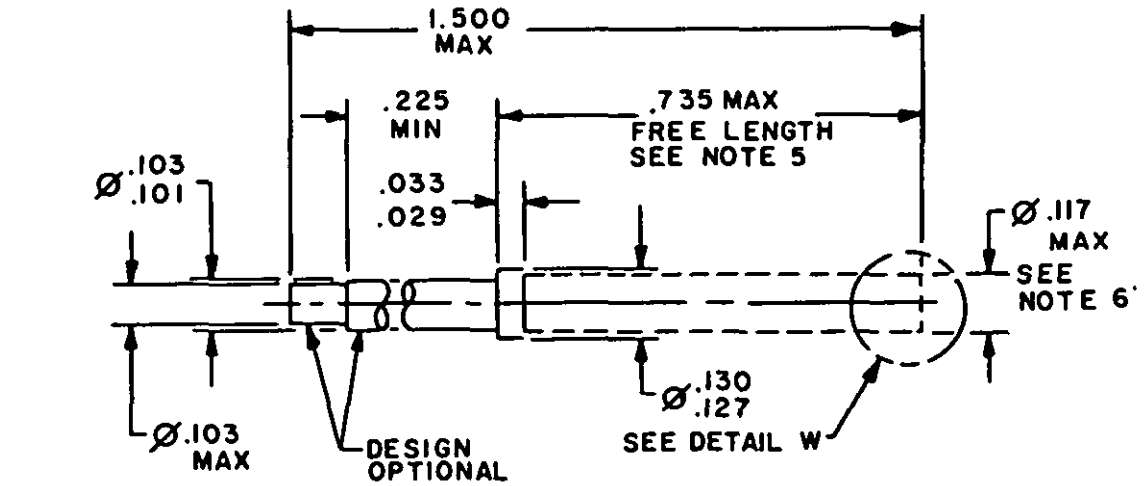
MILITARY SPECIFICATION SHEET

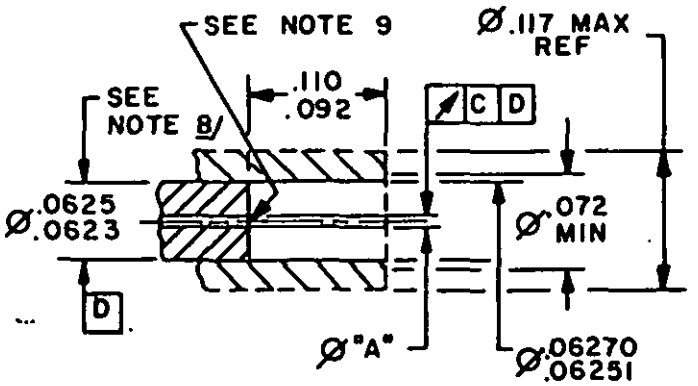
TERMINI, FIBER OPTIC, ENVIRONMENT RESISTING, SOCKET TERMINUS, SIZE 16, REAR RELEASE

MIL-C-38999, SERIES I, III, and IV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the termini described herein consist of this specification and the latest issue of MIL-T-29504.





DETAIL W - DESIGN OPTIONAL
SEE NOTE 7

FIGURE 1. Socket terminus.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026-10-20T23:34:47
Check the source to verify that this is the current version before use.

Inches	mm	Inches	mm
.029	0.74	.103	2.62
.033	0.84	.117	2.97
.072	1.83	.127	3.23
.0625	1.58	.130	3.30
.0627	1.59	.225	5.72
.084	2.13	.575	14.61
.092	2.34	.702	17.83
.100	2.54	1.500	38.10
.101	2.57		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .003 unless otherwise indicated.
4. Dimensions apply after plating when applicable.
5. Free length prior to installation in the connector.
6. Alignment sleeve and body design optional, not to exceed maximum envelope dimensions.
7. For alternate alignment sleeve design, engaging force shall be less than preload spring force (I.D.) tolerance does not apply).
8. Alignment surface O.D. shall be concentric with "A" dia per table I.
9. Polished fiber face may be flat to concave and flush to recess.
10. "A" dia hole tolerance is ±B, see table I.

FIGURE 1. Socket terminus - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:34Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Tools: See table II.

Mating termini: MIL-T-29504/4.

Part number: See figure 1 and table I. Part number M29504/5- (dash number from table I.)

Marking: Part number shall be marked on barrel of the termini.

TABLE I. "A" diameter hole for optical waveguide fiber and part number designation.

TICC	Military part number	"A" diameter		Tolerance		Fiber size	Terminus
		µ	(inch) 1/	Circular runout 2/	Max 1/ B + C (inch)µ	Core/clad µ	Size
4045	M29504/05-4045	124.5±B	(.0049±B)	C			16
4046	M29504/05-4046	127.0±B	(.0050±B)	C	(.00059) 15	50/125	16
4047	M29504/05-4047	129.5±B	(.0051±B)	C			16
4048	M29504/05-4048	139.5±B	(.0055±B)	C			16
4049	M29504/05-4049	142.0±B	(.0056±B)	C	(.00059) 15	100/140	16
4050	M29504/05-4050	145.0±B	(.0057±B)	C			16

1/ Dimensions in parentheses are inches.
2/ As defined by ANSI/ASME Y 14.5M - 1982.

TABLE II. Tools.

End size	Insertion tool	Removal tool
16	M81969/14-03	M81969/14-03

Custodians:
Navy - EC
Air Force - 85
Army - CR

Review activities:
Navy - MC, SH
DLA - ES

User activities:
Navy - OS, YD

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 6060-0026-5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T23:34Z
Check the source to verify that this is the current version before use.